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Special Course in Software Engineering ‘26

Qwen and Python Interaction

Python connection

- Create a new folder and virtual environment in your VS Code or any IDE you are using.
  - Remember the steps you have used so far.
- Then in the IDE terminal, type

```
uv pip install ollama
```

- Then create a small Python file called **qwen_assist.py,**

```
from ollama import chat
response = chat(
    model="qwen3:8b",
    messages=[
        {
            "role": "user",
            "content": "Reply with exactly: Python can talk to Qwen."
        }
    ]
)
print(response.message.content)
```

Python connection (ctd.)

- Then run the file:

```
uv run python qwen_assist.py
```

You should get a response from Ollama

This is similar to what you (may) have done in the last exercise after installing Ollama and Qwen.

Input to AI

Change the previous code to the following

```
from ollama import chat

question = input("Ask your Qwen a question: ")
response = chat(
    model="qwen3:8b",
    messages=[
        {
            "role": "user",
            "content": question
        }
    ]
)
print("\nAI response:")
print(response.message.content)
```

This syntax is necessary to send your question to the **Qwen** model via **Ollama**:

`chat()` : This is a Ollama function, necessary to be used to interact with the target model. In this function, you have to give the name of the model and the actual message (question) as parameters.

`messages`: Stores the list of messages that gets sent to the Ollama and then to Qwen

`role`: Defines the roles of the "**`user`**". When the model replies, the role will be "assistant"

`content`: The actual content that you, as a user, want to pass to the model. Here, it's a question.

The model's response will be stored in **`response`** object.
The actual answer will be available in **`content`**

Prompting AI

Change the previous code to the following

```
from ollama import chat

prompt = "Classify this IT problem: My laptop cannot connect to Wi-Fi."
response = chat(
    model="qwen3:8b",
    messages=[
        {
            "role": "user",
            "content": prompt
        }
    ]
)
print("\nAI response:")
print(response.message.content)
```

Prompting AI (CTD.)

Change the prompt in the previous code to the following:

```
prompt = """
Classify the following IT support problem.

Problem:
My laptop cannot connect to Wi-Fi.

Return exactly ONE of these categories:
NETWORK
SOFTWARE
HARDWARE
ACCOUNT
OTHER

Do not provide an explanation. """
```

Prompting AI (CTD.)

The response from Qwen should be much different from when you had a simpler and vague prompt.

THIS IS THE EFFECT OF GOOD PROMPTING.

What you just did

- You saw few more ways of interacting with your Qwen model, which is hosted on **your** machine
- A brief and quick introduction to **prompting**
- This was just for practice.
- You don't have to submit anything.

That’s all for this practice